

MITCHELL GRAY

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Education

Georgia Tech, Atlanta, GA

Master of Science, Computer Science

GPA: 4.0

Artificial Intelligence

Cornell University, Ithaca, NY

Bachelor of Science, Computer Science

Distributed Systems

Minor, Operations Research Information Engineering

M.Eng Equivalence

Knowledge Domain: Distributed Systems · Systems Programming · AI · Machine Learning · Databases · Cloud

Technical Skills

Languages: C · C++ · Python · Java · Rust · PL/SQL

Software & Tools: Git · Pytorch · Scikit-learn · NumPy · Pandas · Poetry · CMake · CI/CD · Docker · K8s · Azure · AWS

Relevant Work Experience

Oracle

Redwood City, CA

Software Engineer II | C, PL/SQL, Raft, Java, OracleDB, OCI

Sept 2025 – Present

- Working on Raft Replication in the Oracle Globally Distributed Database for Shard Replication · Writing fault-tolerant and scalable Distributed Systems code in **C**, **Java**, and **PL/SQL**.
- Received **Training & Mentorship Excellence Award** · Provided technical mentorship to our newest engineer.
- Designed and implemented a generic shard-operations marker framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new markers with fewer defects.
- Architected and implemented advanced deployment algorithms when specifying server rack requirements to more evenly balance replicas across shards automatically · Improved CLI user-interface to view user-specified rack information.
- Implemented comprehensive Pluggable DBs lifecycle management · Enabled seamless Drop, Unplug, and Plug operations of PDBs to streamline maintenance for Raft replicas using Oracle's multi-tenant container architecture.
- Fixed vulnerabilities in slow-replica flow-control · Implemented new flow-control algorithms to improve fail-over.

Software Engineer I

June 2024 – Sept 2025

- Engineered a fault-tolerant replica replication feature on Oracle Cloud Infrastructure · Enforced placement of Raft replicas on separate server racks to eliminate single-rack replication-unit failures and improve database availability.
- Refactored over 1,200 lines of Sharding serialization code to eliminate a critical data-loss defect · Designed robust de/serialization pipelines and multi-format conversion logic that greatly improved data integrity and system reliability.
- Optimized multiple replica-to-replica data transfer operations by using batching · Achieved a **700%** performance improvement offline and **250%** improvement online after optimization code changes.
- Resolved critical recoverability issue in replica-to-replica data transfer operations by ensuring complete transfer of essential recovery-related tables and data between replicas · Improved reliability and system integrity during failure.
- Enhanced error-handling mechanisms throughout the codebase · Improved error handling reliability and diagnosability.

Gecko Robotics

Pittsburgh, PA

Software Engineer Intern | C++, Python, C, CMake, GCP, CI/CD

May 2023 – Aug 2023

- Worked on the Robot Controls team · Wrote code for an asynchronous distributed system in **C++**, **Python**, and **C**.
- Revamped Robot & DAQ emulators · Implemented communications protocol and Client/Server TCP networking.
- Replaced VS build-system with **CMake** · Added calibration support to emulators · Integrated **CI/CD** and **Poetry**.

Carnegie Mellon University - Software Engineering Institute

Pittsburgh, PA

DevOps Engineer Intern | Python, Rust, CI/CD, Docker, Kubernetes

May 2022 – May 2023

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** live data visualizations for clients · Used ArgoCD and Helm to deploy AWS EKS cluster.
- Built internal analytics tools using **Python**, **Neo4j**, NeoDash, and PageRank-based graph analysis to generate engineering metrics and documentation adopted across multiple teams within the company.

Georgia Tech Contextual Computing Group **Atlanta, GA**
Graduate Researcher | **AI, ML, Python, Typescript** **June 2025 – Aug 2026**

- Worked on Nosi, an IDE that combines an encrypted virtual file system and application behavior-logging · Utilizes **Machine Learning** techniques to analyze behavioral data and flag potentially-cheating students.
- Published at **Learning at Scale 2026** · Integrated the IDE into multiple Georgia Tech graduate-level CS courses.
- Created models that can detect the difference between human-coding and code transcription · Implemented tools to help review flagged cheating cases and replay student interactions with the IDE · Developed scripts that automatically generate encrypted repositories for students and decrypt them for the auto-grader.

Cornell University **Ithaca, NY**
Undergraduate Teaching Assistant | **Python, NumPy, SQL, Java** **Aug 2022 – May 2024**
 • Received the **Course Staff Exceptional Service Award** · Taught Databases, Object-Oriented Programming, and Physics · Lectured 400+ students with custom-designed lecture content · Led Exam sessions and other TAs.

Publications

Learning at Scale (2026) [\[https://dl.acm.org/doi/abs/10.1145/3774398.3811574\]](https://dl.acm.org/doi/abs/10.1145/3774398.3811574)
Demonstrating Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

Awards

Cornell Bowers CIS	<i>Course Staff Exceptional Service Award</i>
Faculty-nominated award for exceptional commitment, hard-work, dedication, and exceptional service of a Teaching Assistant during the 2023-2024 academic year.	

New AI Infra Project Placeholder | Personal Project | *AI, Go* | *Github* May 2026 – Present

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Improving 3D Point-Cloud Classification | Deep Learning | *AI, Python, Pytorch* | *[Github](#)* Aug 2025 – Dec 2025

- Re-implemented and upgraded state-of-the-art point cloud classification Deep Neural Networks · Implemented PointNet and PointPillars · Upgraded these models by modifying their architectures, adding Dropout, implementing Data augmentation, and introducing Residual Connections · Improved validation accuracy by 4% and massively reduced overfitting.

Distributed Testing Platform (SPEED) | Masters Project | *Java* | *Github* | *BOOM* Aug 2023 – May 2024

- 1 of 44 projects selected for BOOM 2024 · Worked on SPEED, Scalable Platform for Efficient Execution of Distributed testing, which is a fault-tolerant system that contains a controller node that orchestrates worker nodes running **JUnit** tests on **Java** code · Aggregated worker node findings to the controller · Displayed test results in the frontend to the user once all tests finish.

SpGEMM Parallelization on GPUs | Parallel Computing | *CUDA, C++* | *Github* Jan 2024 – May 2024

- Parallelized sparse-matrix multiplication operations using GPUs · Experimented with multiple variations of the algorithm which massively improved performance of the algorithms.